

SECTION A (30 Marks)

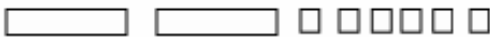
ATTEMPT ALL QUESTIONS

Question 1

Direction: For each question, there are **FOUR** responses: **A, B, C** and **D**. Choose the correct alternative and circle it. **DO not circle more than one alternative. If more than one choice is circled, NO score will be awarded.**

- i. A number in the expanded form is $5 \times + 4 \times + 2 \times$. The number of digits to the right of the decimal point in the standard form is
 - A. 4
 - B. 9
 - C. 15
 - D. 14
- ii. In a class, the ratio of boys to girls is **5: 4**, if there are 500 boys, how many girls are in the class?
 - A. 4 girls
 - B. 250 girls
 - C. 300 girls
 - D. 400 girls
- iii. Which of the following expression represents the pattern rule for the sequence: **4,14,24,34,...**
 - A. $10n$
 - B. $10n - 6$
 - C. $10n + 6$
 - D. $10n + 4$
- iv. The correct scientific notation of **0.021** is
 - A. $21 \times$
 - B. $2.1 \times$
 - C. $2.1 \times$
 - D. $21 \times$

v. Which model below best represent the difference for ?

- A. 
- B. 
- C. 
- D. 

vi. What is the best possible number we can multiply to make perfect square of prime factor:

$$2 \times 2 \times 3 \times 3 \times 3 \times 7 \times 11 \times 11?$$

- A. 2
- B. 3
- C. 7
- D. 11
- vii. “When you input, triple it, and subtract the output is”. The equation of the given statement is
- A.
- B.
- C.
- D.
- viii. Sonam is a student from Mongar Dzongkhag. While shopping, she was confused about which soap to buy because of the different prices shown in the table.

Shops	Quantity	Amounts
A	2 Pieces	Nu. 5
B	5 Pieces	Nu. 10
C	3 Pieces	Nu. 3

Which shop offers the best price per pieces?

- A. Shop A
- B. Shop B
- C. Shop C
- D. All the above options

ix. Apply the order of operation rules to solve $(-5) + (-4) \times (-5) \div 2$. Which of the following is the correct answer?

- A. 1
- B. -3
- C. 2
- D. 5

x. The solution to the equation is

- A.
- B.
- C.
- D.

xi. The table below shows a sequence of numbers.

Term number	Term value
1	3
2	12
3	21
4	30
5	39
6	?

What is the value for next term?

- A. 48
- B. 49
- C. 40
- D. 46

xii. Nima spent $4h$ studying for exam. For each subject, she studied h . How many subjects did she study?

- A. 4 subjects
- B. 5 subjects
- C. 6 subjects
- D. 7 subjects

xiii. A farmer has 120 apples. He decided to sell them at a market. If he sells 40% of the apples in

the morning and the rest in the afternoon, how many apples will he sell in the afternoon?

- A. 48 apples
- B. 72 apples
- C. 60 apples
- D. 50 apples

xiv. . What property of grouping of factors is shown?

- A. closure
- B. associative
- C. distributive
- D. commutative

xv. Lobzang earns a rate of commission of 3%. His commission on a sale of Nu 15,000 is

- A. Nu 1530
- B. Nu 300
- C. Nu 450
- D. Nu 345

SECTION B (50 MARKS)
ATTEMPT ALL QUESTIONS

Question 2.

- a. Rinzin borrows Nu 3500 for 18 months at an annual simple interest rate of 12%. How much will Rinzin have to pay back altogether after 18 months? **[3]**

- b. How do you know that the situation described below is a linear relationship?

The distance travelled by a plane flying at 550 km / h compared to the numbers of hours flying **[2]**

Question 3

- a. Write each number in standard form and compare which number is greater? [3]

$$7 \times 10^{-1} + 3 \times 10^{-2} + 1 \times 10^{-4} \text{ and } 6 \times 10^0 + 3 \times 10^{-1} + 3 \times 10^{-2} + 2 \times 10^{-3}$$

- b. A tank contains 400 L of water. It is emptied at a rate of 32 L/min. Write an equation you could solve to determine how long it will take until the tank contains only 112 L of water. Solve the equation. [2]

Question 4

- a. The area of Yeshay's chessboard top is 64 . Her friend Wangmo wanted to make a chessboard of the same size. What would be the side length of Wangmo's chessboard? [2]

- b. The diameter of a red blood cell is **0.0000074cm** and the diameter of COVID-19 virus is **cm**. Express both diameters in scientific notation and compare their sizes.

[3]

Question 5

- a. The square field has an area of **3600 m²**. What is its perimeter? [3]

- b. A shopkeeper sold a mobile phone at Nu 12,000 during Mongar Tshechu. The actual price was Nu.15,000 in the shop. What was the discount percent? [2]

Question 6

- a. The ratio of the length of the photo to the length of the photo frame as shown in the figure below is **2: 3**. If the length of the photo frame is 36 cm, what will be the length of the photo?

[2]



- b. Dawa Dem has saved Nu 150. Each month, she will add Nu 25 to her savings.

What amount will be available in her account after 10th month, if she doesn't withdraw?

[3]

Question 7

- a. One evening, the temperature started at 5°C. If the temperature dropped by 17°C in the next six hours, what will be the new temperature?

[2]

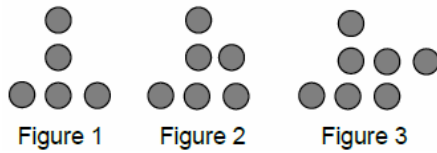
- b. Tashi Planted 5 seedlings for every 3 seedlings Dorji planted. Dorji planted 90 seedlings in 1h.
- i. How many seedlings did Tashi plant in 1 h? [1]

- ii. How long did it take Dorji to plant 30 seedlings?
[1]

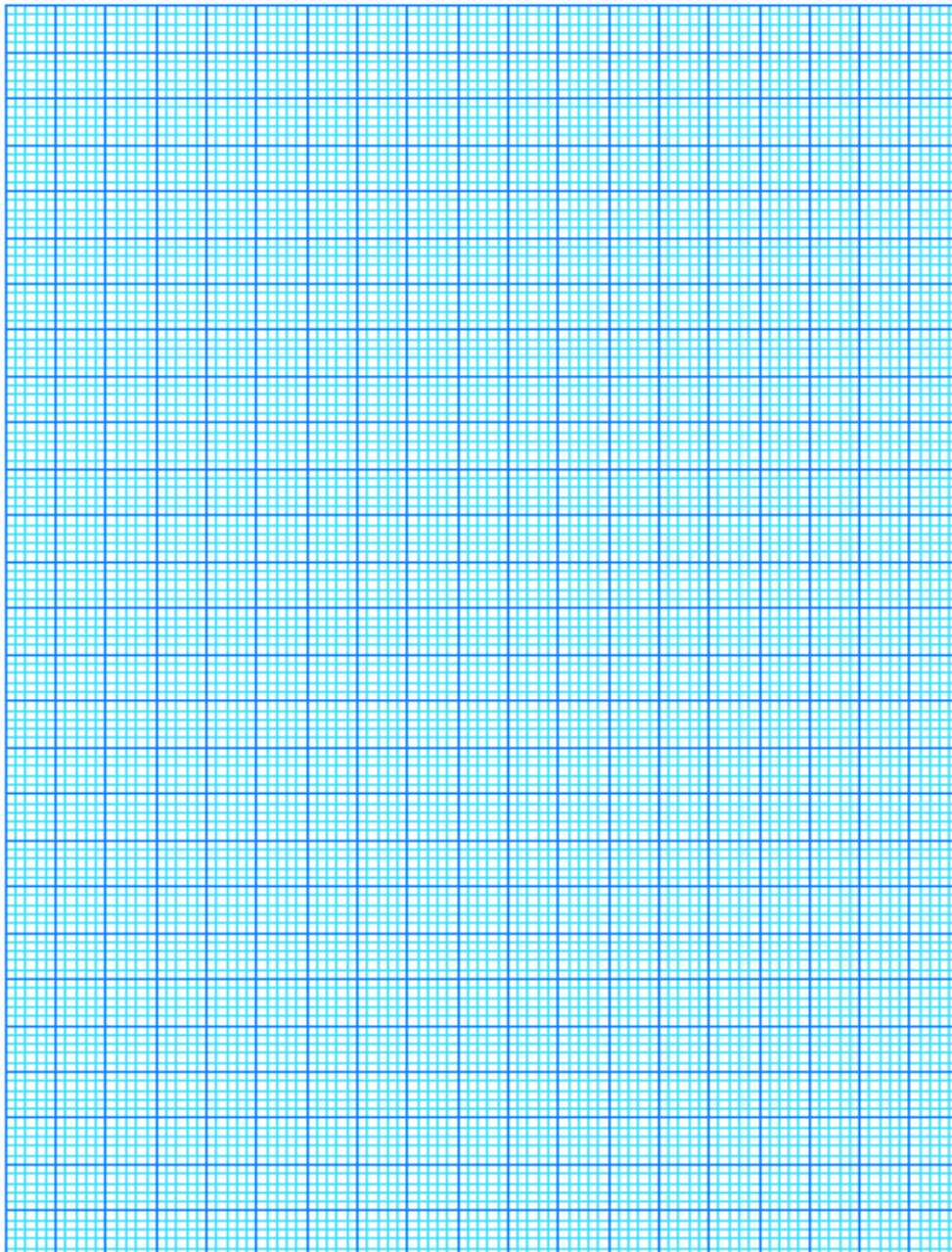
- iii. How long did it take Tashi to plant 30 seedlings?
[1]

Question 8

- a. Examine the pattern below.



- i. Draw a graph to show the relationship between the figure number and the number of dots in that figure. [1]
- ii. Use your graph to predict the number of dots in Figure 10. [1]



- b. List a pair of possible polynomials for each. [3]
- i. the sum of the pair is $-4x - 7$
 - ii. the difference of the pair is $-x - 5$

Question 9

- a. The number is written as:
- i. How many digits are there to the right of the decimal point if you write this number in standard form? Explain how you know. [2]
 - ii. Could you write this number using digits after the decimal point? Justify your answer. [1]

- b. Dawa has read $\frac{1}{4}$ of a book. He has read 15 pages. How many pages are there in the whole book? [2]

Question 10

- a. Ugyen and his friends need $1\frac{1}{4}$ buckets of paint to white-wash one side of their classroom's wall. How much paint will they need to paint one fifth of the same wall? [3]

- b. The table below shows a speed of sound in air and water

Medium	Speed in (m/sec)
Air	
water	

- i. Which medium is the speed of sound faster? [1]
ii. What is the difference in speed? [1]

Question 11

- a. i. How many are there in ? [1]
ii. Draw a picture to show why your answer makes sense. [2]

- b. What happens when you multiply any integer except 0 by -1 ? [2]

THE END

(ROUGH)